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EXAMPLE-MODERN-TODAY-BRING-ME-NOT-ALL-HIGH-BODIES

K-MEANS CLUSTERING

The city officials want to identify neighborhoods or zones that are most stressed, so they can establish mental health support centers.













































































C1



C2

Each data point is a geolocation of a survey response, possibly weighted by the stress level reported. Using K-Means, the city is divided into K clusters. The centroids of clusters with the highest average stress scores could be the recommended locations for the mental health support centers.



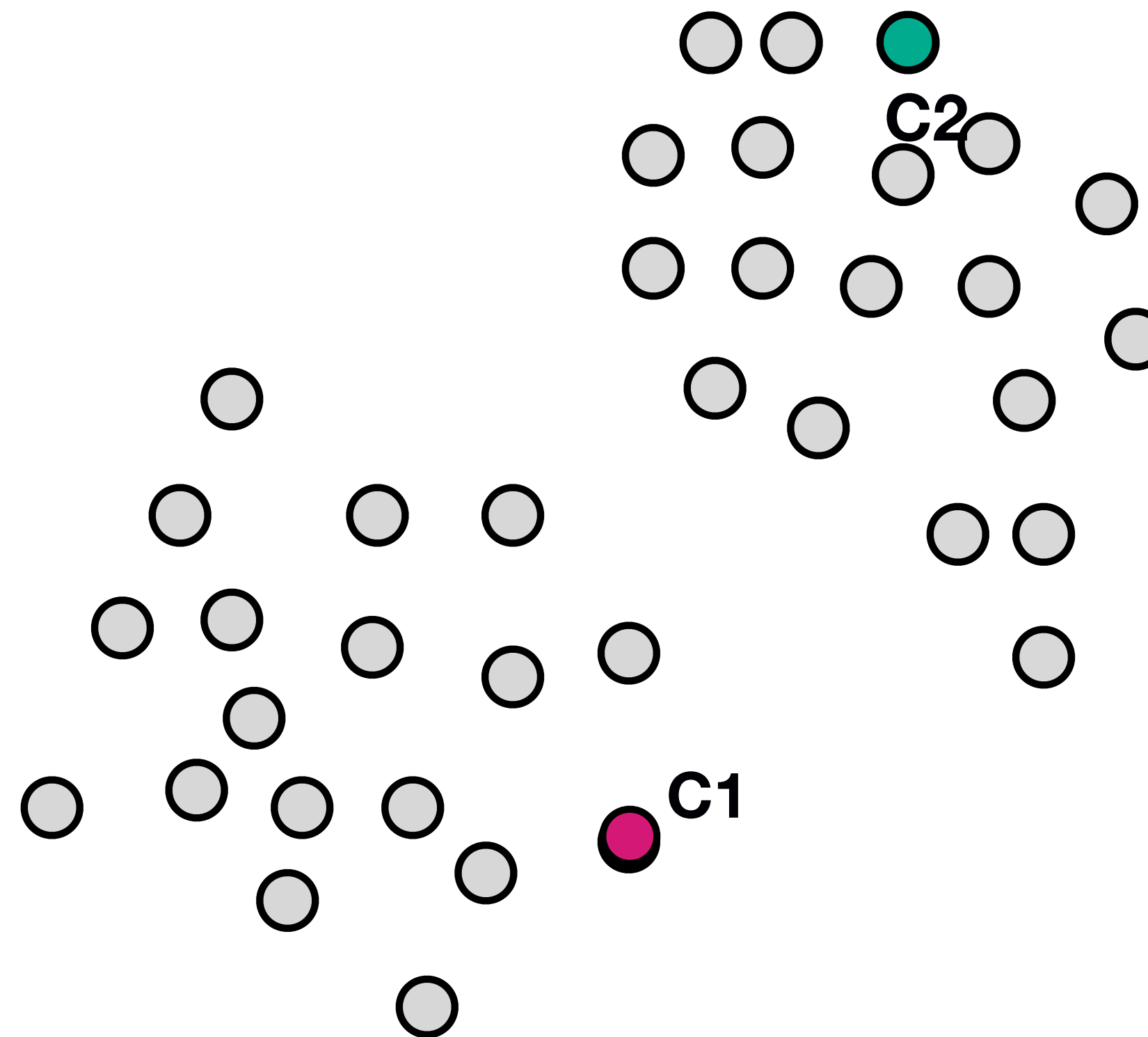
c1

K-MEANS CLUSTERING

EXAMPLE-MONITORING MENTAL HEALTH IN DIFFERENT NEIGHBORHOODS

The city officials want to identify neighbourhoods or zones that are most stressed, so they can establish mental health support centers.

Each data point is a geolocation of a survey response, possibly weighted by the stress level reported. Using K-Means, the city is divided into K clusters. The centroids of clusters with the highest average stress scores could be the recommended locations for the mental health support centers.



K-MEANS CLUSTERING

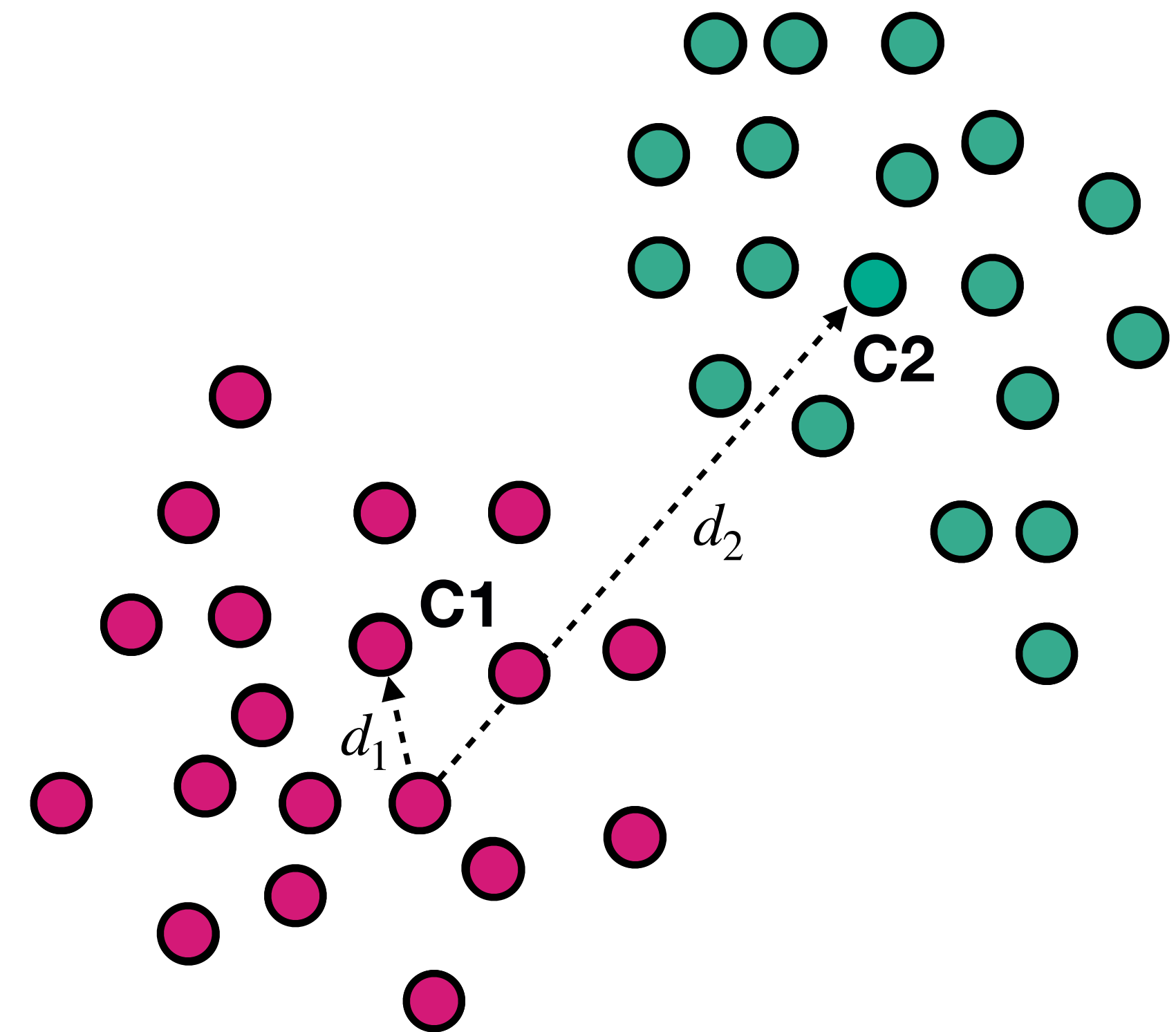
FROM NEIGHBOURHOODS TO GROUPING!

Objective– Partition n data points into K clusters such that the total within-cluster sum of square (WCSS) is minimized. Formally, the objective is to find:

$$\arg \min_S \sum_{i=1}^k \sum_{x \in S_i} ||x - \mu_i||^2^*$$

Where:

- S are the K clusters.
- μ_i is the centroid or mean of all the points in S_i .



* The $|| \cdot ||$ notation in formulas often denotes a norm, which is a function that assigns a positive length or size to each vector in a vector space, excluding the zero vector which is always assigned a length of zero.